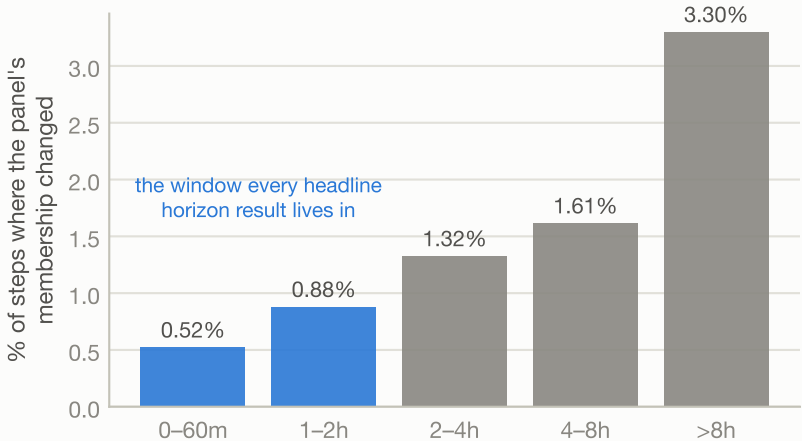


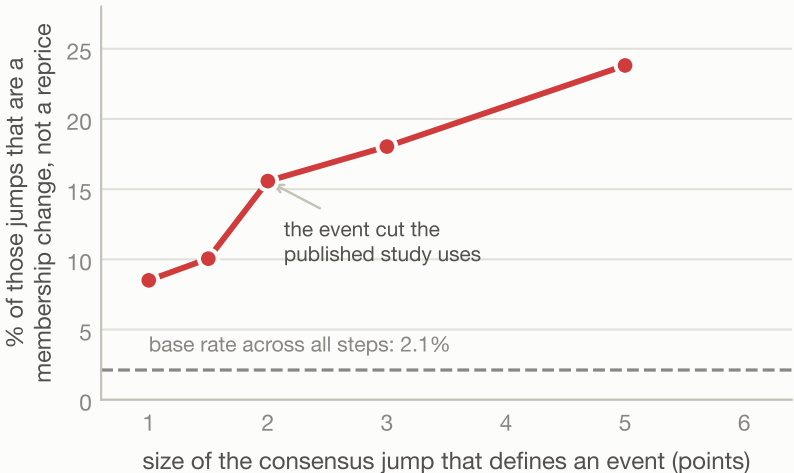
Why the benchmark had to be audited before its timing could be trusted

"The sportsbook" in every lead-lag result is a mean across books. When a book joins or drops out, the mean moves without anyone repricing — and a jump-shaped artifact lands in a study whose events are jumps.

The churn lives far from kickoff



and it concentrates in exactly the big moves



book_panel.log \$1-\$2 · 15-minute grid. Membership change is detected as a change in book count, a LOWER bound: one book leaving as another joins is invisible to it.